

Rutgers-The State University
Master's Comprehensive Examination

Part I

Theory

December 7, 1985

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
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- 1) k balls are randomly placed into n boxes. Find the expected value and the variance of the number of empty boxes.
(Hint: Use indicator functions.)
2. Let $X_1, \dots, X_n$ be i.i.d. Poisson (λ).
 - i) Find the UMVU estimator of $e^{-a\lambda}$.
 - ii) Show that if $a > n$ the estimator in i) is poor.
3. Let X be the only observation from the density $f(x; \theta) = (\theta + 1)x^\theta I_{(0,1)}(x)$, where $\theta > -1$.
 - (i) In testing $H_0: \theta \leq 1$ vs $H_1: \theta > 1$, find the power function and size of the following test: reject H_0 iff $X > \frac{1}{2}$.
 - (ii) Find a uniformly most powerful size $-\alpha$ test of $H_0: \theta \geq 2$ vs $H_1: \theta < 2$.
4. Let $X \sim B(n; p)$ and the prior distribution of p is Beta (c, c).
 - i) Find the Bayes estimator of p if loss is squared error.
 - ii) Find the risk of the estimator in i).
 - iii) Find c so that the risk is a constant.
5. Let $X_1, \dots, X_n$ be i.i.d. from the following distribution. What is the maximum likelihood estimator of θ .

$$f(x) = (\theta + 1)x^\theta I_{(0,1)}(x) \text{ for } \theta > -1.$$

Is the estimator consistent with $\theta > -1$. Estimate θ by moments.

6. Let X be the number of successes in n independent trials with probability p .
 - i) Find the UMP test for testing $p \leq p_0$ vs. $p > p_0$ for $n = 6$, $p_0 = .25$ and $\alpha = .1$.
 - ii) Find the power of the test against $p_1 = .3, .5$.
 - iii) If $p_0 = .2$ and $\alpha = .05$, and it is desired to have power $\beta > .9$ against the alternative $p_1 = .4$, find the necessary sample size by the normal approximation.

Rutgers-The State University
Master's Comprehensive Examination

December 7, 1985

Applied

Part II

- Instructions:
1. Open book and open notes
 2. Do four out of the following five problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

1. Three different washing solutions are being compared to study their effectiveness in retarding bacteria growth in 5-gallon milk containers. The analysis is done in a laboratory, and only three trials can be run on any day. On each day the 3 solutions are randomly prepared. Analyze the data and draw conclusions.

Solution	Days				ΣX_i	ΣX_i^2
	1	2	3	4		
1	13	22	18	39	92	2498
2	16	24	17	44	101	3057
3	5	4	1	22	32	526
ΣX_i	34	50	36	105	225	
ΣX_i^2	450	1076	614	3941		

- (a) What type of design is this? What is the model and assumptions?
 - (b) How would you check the assumptions of the analysis?
 - (c) Test for no differences in mean effectiveness of the 3 solutions.
 - (d) Test for a difference between solution 1 and solution 3 using a post hoc (i.e. multiple comparison) procedure.
 - (e) Assuming that chemical types and bolts are fixed, estimate the model parameters.
2. (a) The hour of birth has been recorded for all the births in a year in a maternity hospital. Commencing at the hour 12 midnight to 1 a.m. the results are given in the following table for the first six hours.
- Number of births - 97, 136, 151, 110, 144, 136. Test the null hypothesis that births occur uniformly in the first 6 hours of the day. Give the P-value. Comments?
- (b) A hospital administrator wants to determine whether the number of hour arrivals in the outpatient department can be described by a Poisson distribution (Work schedules of hospital employees may be constructed with the help of such knowledge.) A simple random sample of 50 hours reveals the data of Table 1.

Table I

Number of Outpatient Arrivals per Hour	Observed Frequency, f_0
0	0
1	1
2	5
3	8
4	15
5	9
6 or more	12
Total	50

Note: Prior experience gives the mean number of arrivals at 3.8.

- (b) Test the null hypothesis that the number of births in each hour follows a Poisson distribution. Give the P-value. Comments?

3. A substance used in biological and medical research is shipped by air freight in cartons of 1,000 ampules. The data below, involving eight shipments, were collected on the number of times the carton was transferred from one aircraft to another (including the initial loading) over the shipment route (X) and the number of ampules found to be broken upon arrival (Y).

i:	1	2	3	4	5	6	7	8
X_i :	2	1	3	1	4	2	1	2
Y_i :	16	9	17	12	22	13	8	15

- (a) Obtain the estimated regression equation. Does a linear regression function appear to give a good fit here? Why?
- (b) Give a 99% C.I. for the slope.
- (c) Obtain a point estimate of σ^2 . In what units is your estimate expressed?
- (d) Give a prediction and 95% C.I. for a new observation at $x = 3$.

4. In the table below mothers of children who had been referred by their teachers as presenting behavioral problems and compared with mothers of children who had not been so referred. The two groups are described respectively as problems and controls. A record was made for each mother whether or not she had suffered any infant loss (e.g. stillbirths) prior to the birth of the child in the study.

Numbers of mothers with	Birth Order			
	2		3-4	
	Problem controls		Problem controls	
Losses	20	10	26	16
None	82	54	41	30
Percent loss	19.6	15.6	38.8	34.8

- (a) Test for independence among the 3 categories, give the P-value.
- (b) Test for independence among the 2 categories at birth order 3-4. Give the P-value. Explain the discrepancy with part (a) above.
5. Consider the following data which gives miles per gallon of a make of car before and after application of a newly developed gasoline additive.
- (a) Test the null hypothesis of no effect of this new gasoline additive (use a nonparametric method).
- (b) Estimate the effect of this new gasoline additive and give a 95% confidence interval.

Car	1	2	3	4	5	6	7
Before	17.2	21.6	19.5	19.1	22.0	18.7	20.3
After	18.3	20.8	20.9	21.2	22.7	18.6	21.9

Rutgers-The State University
Master's Comprehensive Examination

March 9, 1985

Theory

Part I

- Instructions:
1. Open book and open notes
 2. Do three out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

1. Let $X_1, \dots, X_n$ be i.i.d. with density $f(x, \theta) = (2\theta+1)^{-1} \exp\{-\frac{x}{2\theta+1}\}$ $x > 0$, $\theta > -1/2$.
 - a) Find the m.l.e. $\hat{\theta}$ of θ .
 - b) Show that the m.l.e. is unbiased.
 - c) Find the m.s.e. of $\hat{\theta}$. Can you find another unbiased estimate of θ which has a smaller m.s.e. than the m.l.e.?
2. Let $X_1, \dots, X_n$ be i.i.d $N(1-2\theta, 1)$,
 - (a) Find the UMP test at level α for $H_0: \theta = 0$ vs $H_1: \theta > 0$.
 - (b) For $\alpha = .05$ and $n = 25$, find the critical region.
 - (c) For $\alpha = .05$ and $n = 25$ calculate the power of the test at $\theta = .1$.
3. Let (X, Y) have the joint density $f(x, y, \theta) = \frac{12\theta^3}{y^5}$, $0 < \theta < x < y < \infty$. θ is the unknown parameter to be estimated.
 - (a) Find a sufficient statistics for θ .
 - (b) Show that $d(Y) = Y/2$ is an unbiased estimate of θ .
 - (c) Find an unbiased estimate of θ which has smaller m.s.e. than the estimate in (b).
4. The lifetime (in number of hours) X of a component has the following continuous density $f(x) = \frac{1}{2(1+x)^{3/2}}$ $x > 0$. We will take a random sample of n components with the above density. We want that n should be large enough so that with probability at least .95 between 24% and 26% of the sample components last at least fifteen hours. Find the smallest value of n using
 - (a) Chebyshev's inequality
 - (b) Central limit theorem.

[Note that $\int_0^t f(u) du = 1 - \frac{1}{(1+t)^{1/2}}$ for any $t > 0$].

5. Let X have the density $f(x|\theta) = \frac{1}{\theta} 0 < x < \theta, \theta > 1$. The prior density of θ is $\pi(\theta) = \frac{3}{\theta^4} \theta > 1$.
- (a) Find the Bayes estimate of θ under the squared error loss.
 - (b) Is the Bayes' estimate in (a) unbiased?
 - (c) Find the Bayes' estimate of θ if the loss function is $L(\theta, a) = \frac{(a-\theta)^2}{\theta^2}$.
6. A box contains 3 red and 2 white balls. Four balls are chosen, at random, without replacement from the box.
- (a) What is the probability that you will get at least 2 white balls before you get any red balls?
 - (b) What is the expected number of white balls you get before your first chosen red ball?
 - (c) If the total number of white balls chosen is 2 what is the probability that the first ball was red?